

Total Items	24								
Pass Items	18								
Fail Items	6								
NY Items									
Section	Main Title	Description	Testname	Priority	Method	Owner	Status	Milest one	Remark
1	Register test								
1.1	Read default value	1. Wait for reset and clock stable 2. Read value of 4 register Pass condition: Check value matching with default value mention in RTL spec	reg_def_test	High	Directed	Minh Phan	PASSED		
1.2	Read and Write value check	1. Wait for reset and clock stable 2. Write random value of 4 register 3. Read value of 4 register Pass condition: Check value matching	reg_rw_test		Directed	Minh Phan	PASSED		
1.3	Reset on the fly check	1. Wait for reset and clock stable 2. Write random value of 4 register 3. Reset system 4. Read value of 4 register Pass condition: Check value matching with default value mention in RTL spec	reg_rs_test		Directed	Minh Phan	PASSED		
1.4	Check reserved register	1. Wait for reset and clock stable 2. Write random value of reserved register 3. Read value of reserved register Pass condition: Check value equal is 0	reg_reserved_test		Directed	Minh Phan	PASSED		
1.5	Check interrupt in TSR register	1. Wait for reset and clock stable 2. Write data 8'b11 to TIE register 3. Write data 8'h01 to TCR register 4. Wait interrupt is asserted 5. Write data 8'h00 to TSR register 6. Read value of TSR register 7. Write data 8'b01 to TSR register 8. Read value of TSR register 9. Write data 8'b11 to TCR register 10. Wait interrupt is asserted 11. Write data 8'b00 to TSR register 12. Read value of TSR register 13. Write data 8'b10 to TSR register 14. Read value of TSR register Pass condition: Check underflow bit and overflow bit is	reg_tsr_test		Directed	Minh Phan	PASSED		
2	Counter combine with divisor test								
2.1	Count up								
2.1.1	Count from 0, no divide	1. Wait for reset and clock stable 2. Write data 8'b01 to TIE register 3. Write data 8'b01 to TCR register 4. Wait interrupt trigger 5. Read value of TSR register Pass condition: Check interrupt is asserted and read value is 8'h01	cnt_up_div1_test	High	Directed	Minh Phan	PASSED		

Section	Main Title	Description	Testname	Priority	Method	Owner	Status	Milest one	Remark
2.2.1	Count from 255, no divide	1. Wait for reset and clock stable 2. Write data 8'hFF to TDR register 3. Write data 8'b000_00_100 to TCR register 4. Write data 8'b10 to TIE register 5. Write data 8'h02 to TCR register 6. Write data 8'h03 to TCR register 7. Wait interrupt asserted 8. Read value of TSR register Pass condition: Check interrupt is asserted and read value is 8'h02	cnt_dw_div1_test	High	Directed	Minh Phan	PASSED		
2.2.2	Count from 255, divide by 2	1. Wait for reset and clock stable 2. Write data 8'hFF to TDR register 3. Write data 8'b000_00_100 to TCR register 4. Write data 8'h02 to TIE register 5. Write data 8'h02 to TCR register // set count down 6. Write data 8'b000_01_011 to TCR register // star count down with devide by 2 7. Wait interrupt asserted 8. Read value of TSR register Pass condition: Check interrupt is asserted and read value is 8'h02	cnt_dw_div2_test	Medium	Directed	Minh Phan	PASSED		
2.2.3	Count from 255, divide by 4	1. Wait for reset and clock stable 2. Write data 8'hFF to TDR register 3. Write data 8'b000_00_100 to TCR register 4. Write data 8'h02 to TIE register 5. Write data 8'h02 to TCR register // set count down 6. Write data 8'b000_10_011 to TCR register // star count down with devide by 4 7. Wait interrupt asserted 8. Read value of TSR register Pass condition: Check interrupt is asserted and read value is 8'h02	cnt_dw_div4_test		Directed	Minh Phan	PASSED		The underflow is NOT LOW after write 8'h02 to TSR_ADDR
2.2.4	Count from 255, divide by 8	1. Wait for reset and clock stable 2. Write data 8'hFF to TDR register 3. Write data 8'b000_00_100 to TCR register 4. Write data 8'h02 to TIE register 5. Write data 8'h02 to TCR register // set count down 6. Write data 8'b000_11_011 to TCR register // star count down with devide by 8 7. Wait interrupt asserted 8. Read value of TSR register Pass condition: Check interrupt is asserted and read value is 8'h02	cnt_dw_div8_test		Directed	Minh Phan	PASSED		The underflow is NOT LOW after write 8'h02 to TSR_ADDR
2.2.5	Count from random, no devide	1. Wait for reset and clock stable 2. Write random value to TDR register 3. Write data 8'b000_00_100 to TCR register 4. Write data 8'h02 to TIE register 5. Write data 8'h02 to TCR register // set count down 6. Write data 8'h03 to TCR register // star count down with devide by 8 7. Wait interrupt asserted 8. Read value of TSR register Pass condition: Check interrupt is asserted and read value is 8'h02	cnt_up_rd_div1_test		Random	Minh Phan	PASSED		

Section	Main Title	Description	Testname	Priority	Method	Owner	Status	Milest one	Remark
2.2.6	Count from random, devide by 2	1. Wait for reset and clock stable 2. Write random value to TDR register 3. Write data 8'b000_00_100 to TCR register 4. Write data 8'h02 to TIE register 5. Write data 8'h02 to TCR register // set count down 5. Write data 8'b000_01_011 to TCR register 4. Wait interrupt trigger 5. Read value of TDR register Pass condition: Check interrupt is asserted and read value is 8'h02	cnt_up_rd_div2_test		Random	Minh Phan	PASSED		
2.2.7	Count from random, devide by 4	1. Wait for reset and clock stable 2. Write random value to TDR register 3. Write data 8'b000_00_100 to TCR register 4. Write data 8'h02 to TIE register 5. Write data 8'h02 to TCR register // set count down 5. Write data 8'b000_10_011 to TCR register 4. Wait interrupt trigger 5. Read value of TDR register Pass condition: Check interrupt is asserted and read value is 8'h02	cnt_up_rd_div4_test		Random	Minh Phan	PASSED		The underflow is NOT LOW after write 8'h02 to TSR_ADDR
2.2.8	Count from random, devide by 8	1. Wait for reset and clock stable 2. Write random value to TDR register 3. Write data 8'b000_00_100 to TCR register 4. Write data 8'h02 to TIE register 5. Write data 8'h02 to TCR register // set count down 5. Write data 8'b000_10_011 to TCR register 4. Wait interrupt trigger 5. Read value of TDR register Pass condition: Check interrupt is asserted and read value is 8'h02	cnt_up_rd_div8_test		Random	Minh Phan	PASSED		The underflow is NOT LOW after write 8'h02 to TSR_ADDR
2.3	Cross count up and count down								
2.3.1	Count up from 0, then changes to count down	1. Wait for reset and clock stable 2. Write data 8'h03 to TIE register 3. Write data 8'h01 to TCR register 4. Repeat 200 ker_clk 5. Write data 8'h02 to TCR register 6. Write data 8'h03 to TCR register 7. Read value of TSR register Pass condition: Check interrupt is asserted and read value is 8'h02	cnt_up_change_test		Random	Minh Phan	PASSED		
2.3.2	Count down from 255, then changes count up	1. Wait for reset and clock stable 2. Write data 8'hFF to TDR register 3. Write data 8'b000_00_100 to TCR register 4. Write data 8'h03 to TIE register 5. Write data 8'h02 to TCR register // set count down 6. Write data 8'h03 to TCR register 7. Repeat 200 ker_clk 8. Write data 8'h01 to TCR register 9. Read value of TSR register Pass condition: Check interrupt is asserted and read value is 8'h01	cnt_down_change_test		Random	Minh Phan	PASSED		
2.4	Changes divide mode at middle								

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2.4.1	Count up with no divide, then random changes divide	1. Wait for reset and clock stable 2. Write data 8'h01 to TIE register 3. Write data 8'h01 to TCR register 4. Repeat 50 ker_clk 5. Random rand_div 5. Write data tcr_val to TCR register 6. Wait interrupt asserted 7. Read value of TSR register Pass condition: Check interrupt is asserted and read value is 8'b01	cnt_up_rd_change_div_test	Medium	Directed	Minh Phan	PASSED		
2.4.2	Count down with no divide, then random changes divide	1. Wait for reset and clock stable 2. Write data 8'hFF to TDR register 3. Write data 8'b000_00_100 to TCR register 4. Write data 8'h02 to TIE register 5. Write data 8'h02 to TCR register 6. Write data 8'h03 to TCR register 4. Repeat 50 ker_clk 5. Random rand_div 5. Write data tcr_val to TCR register 6. Wait interrupt asserted 7. Read value of TSR register Pass condition: Check interrupt is asserted and read value is 8'b02	cnt_down_rd_change_div_test	Medium	Directed	Minh Phan	PASSED		
3 Interrupt test									
3.1	Check underflow forward input	1. Wait for reset and clock stable 2. Write data 8'h02 to TIE register 3. Write data 8'hFF to TDR register 4. Write data 8'b000_00_100 to TCR register 5. Write data 8'h02 to TCR register 6. Write data 8'h03 to TCR register 7. Wait interrupt asserted 8. Read data value from TSR register Pass condition: Check interrupt is asserted	underflow_en_test	High	Directed	Minh Phan	PASSED		
3.2	Check underflow not forward input	1. Wait for reset and clock stable 2. Write data 8'hFF to TDR register 3. Write data 8'b000_00_100 to TCR register 4. Write data 8'h02 to TCR register 5. Write data 8'h03 to TCR register 6. Wait interrupt asserted 7. Read data value from TSR register Pass condition: Check interrupt is NOT asserted	underflow_dis_test		Directed	Minh Phan	PASSED		
3.3	Check overflow forward input	1. Wait for reset and clock stable 2. Write data 8'h01 to TIE register 3. Write data 8'h01 to TCR register 4. Wait interrupt asserted 5. Read data value from TSR register Pass condition: Check interrupt is asserted	overflow_en_test		Directed	Minh Phan	PASSED		
3.4	Check overflow not forward input	1. Wait for reset and clock stable 2. Write data 8'h01 to TCR register 3. Wait interrupt asserted 4. Read data value from TSR register Pass condition: Check interrupt is NOT asserted	overflow_dis_test		Directed	Minh Phan	PASSED		
4 Check APB protocol					Assertion				